

Кросс-компиляция

Собрать пакет openssh и все библиотеки от которых он зависит под архитектуру arm. Собранный пакет добавить в корневую файловую систему из предыдущего задания, запустить в qemu и проверить работоспособность. В качестве отчета подготовить документ с командами по сборке библиотек и самого пакета, привести также команды по добавлению пакета и библиотек в корневую файловую систему и добавить скриншоты работы утилиты ssh в qemu.

1)

Скачиваем нормальный ARM toolchain от Linaro:

```
wget
```

```
https://releases.linaro.org/components/toolchain/binaries/latest-7/arm-linux-gnueabi/gcc-linaro-7.5.0-2019.12-x86_64_arm-linux-gnueabi.tar.xz
```

Распаковала, добавила в PATH:

```
tar xf gcc-linaro-7.5.0-2019.12-x86_64_arm-linux-gnueabi.tar.xz
```

```
export PATH=$PWD/gcc-linaro-7.5.0-2019.12-x86_64_arm-linux-gnueabi/bin:$PATH
```

Скачала openssh:

```
wget https://cdn.openbsd.org/pub/OpenBSD/OpenSSH/portable/openssh-9.7p1.tar.gz
```

распаковала

```
./openssh-9.7p1 develop *1 !12 ?30 > nano INSTALL
```

Смотрим, от каких версий библиотек зависит:

```
The remaining items are optional.

A working installation of zlib:
Zlib 1.1.4 or 1.2.1.2 or greater (earlier 1.2.x versions have problems):
https://zlib.net/

libcrypto from either of LibreSSL or OpenSSL. Building without libcrypto
is supported but severely restricts the available ciphers and algorithms.
```

Zlib выше 1.2.1.2 !!!!!!!!!!!!!!!

```
LibreSSL/OpenSSL should be compiled as a position-independent library
(i.e. -fPIC, eg by configuring OpenSSL as "./config [options] -fPIC"
or LibreSSL as "CFLAGS=-fPIC ./configure") otherwise OpenSSH will not
be able to link with it. If you must use a non-position-independent
libcrypto, then you may need to configure OpenSSH --without-pie.

If you build either from source, running the OpenSSL self-test ("make
tests") or the LibreSSL equivalent ("make check") and ensuring that all
tests pass is strongly recommended.
```

OpenSSL возьмем

Остальные зависимости, что там перечислены не нужны для задания

скачаем **zlib** версии **1.3.1**: `wget https://zlib.net/zlib-1.3.1.tar.gz`

Распакуем `tar -xvf zlib-1.3.1.tar.gz`

`cd zlib-1.3.1`

Подготовим сборку, указывая компилятор под arm

`CC=arm-linux-gnueabi-gcc make`

`make install`

у нас появились заголовочные файлы и собранные библиотеки:

```
> cd _install
> ls
include lib share
> cd lib
> ls
libz.a libz.so libz.so.1 libz.so.1.3.1 pkgconfig
> cd ..
> cd include
> ls
zconf.h zlib.h
```

из-за выработанной паранойи проверим:

```
> file libz.so.1.3.1
libz.so.1.3.1: ELF 32-bit LSB shared object, ARM, EABI5 v
cally linked, BuildID[sha1]=0cd5a28e4b2d893020d545c2cacff
ed
```

оке, arm

Для удобства скопируем (необязательно, но просто)

/m4	/contrib	.skipped-commit-ids
Создать новый каталог		
Введите имя каталога:		
zlib		
[< Дальше >] [Прервать]		

Копирование	
Копировать эти каталоги (3 шт.) с исходным шаблоном:	
* [x] Метасимволы shell	
В:	
d/src/openssh-9.7p1/zlib	
[] Разыменовывать ссылки	[] Внутри подкаталога, если есть
[x] Сохранять атрибуты	[] Изменять относительные ссылки
[< Дальше >] [В фоне] [Прервать]	

Теперь OpenSSL:

```
wget https://www.openssl.org/source/openssl-1.1.1w.tar.gz
```

```
tar -xvf openssl-1.1.1w.tar.gz
```

```
cd ../openssl-1.1.1w
```

Экспортируем тулчейн:

```
export CC=arm-linux-gnueabi-gcc
```

```
export AR=arm-linux-gnueabi-ar
```

```
export RANLIB=arm-linux-gnueabi-ranlib
```

говорим openssl под какую архитектуру собирать + в INSTALL было сказано, что openssl should be compiled as position independent, поэтому fPIC

```
./Configure linux-armv4 shared --prefix=$PWD/_install -fPIC
```

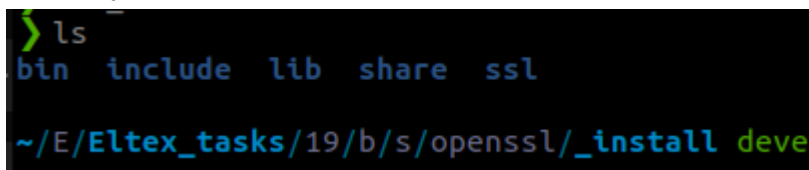
А еще я до этого случайно собрала статически openssl, а всё остальное динамически, и потом думала часами, что же не так

Собираем openssl:

```
make -j12 (у меня 12 ядер)
```

```
make install
```

В итоге у нас:



```
> ls  
bin include lib share ssl  
~/E/Eltex_tasks/19/b/s/openssl/_install devel
```

скопировала содержимое _install в новую папку openssl в openssl

Вернемся к OpenSSH

```
cd openssl-9.7p1
```

добавляем include-пути для компилятора :

```
export CFLAGS="-I$PWD/zlib/include -I$PWD/openssl/include"
```

и пути к .so библиотекам:

```
export LDFLAGS="-L$PWD/zlib/lib -L$PWD/openssl/lib"
```

(иначе configure ругался)

попытка 4

Конфигурируем с нашими зависимостями:

```
./configure --prefix=$PWD/_install --host=arm-linux-gnueabi --with-zlib=$PWD/zlib
```

```
--with-ssl-dir=$PWD/openssl --with-privsep-user=nobody --disable-strip
```

--with-privsep-user=nobody указывает, каким пользователем будет работать непривилегированный процесс sshd, когда включена Privilege Separation (процесс sshd сначала запускается от root
потом создаёт низкопривилегированный процесс для обработки сети, он не может работать от root, ему нужен пользователь nobody) если не указать, sshd при запуске напишет Privilege separation user sshd does not exist
поэтому потом надо будет потом создать юзера

```
OpenSSH has been configured with the following options:
    User binaries: /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/bin
    System binaries: /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/sbin
    Configuration files: /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/etc
    Askpass program: /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/libexec/ssh-askpass
    Manual pages: /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/share/man/manX
    PID file: /var/run
    Privilege separation chroot path: /var/empty
    sshd default user PATH: /usr/bin:/bin:/usr/sbin:/sbin:/home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/bin
    Manpage format: doc
    PAM support: no
    OSF SIA support: no
    KerberosV support: no
    SELinux support: no
    libedit support: no
    libidns support: no
    Solaris process contract support: no
    Solaris project support: no
    Solaris privilege support: no
    IP address in $DISPLAY hack: no
    Translate v4 in v6 hack: yes
    BSD Auth support: no
    Random number source: OpenSSL internal ONLY
    Privsep sandbox style: seccomp_filter
    PKCS#11 support: yes
    UZF/FIDO support: yes

Host: arm-unknown-linux-gnueabihf
Compiler: arm-linux-gnueabihf-gcc
Compiler flags: -I/home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/zlib/include -I/home/alyssa/Eltex/Eltex_tasks/19_cross_c
-pipe -Wno-error=format-truncation -Wall -Wextra -Wpointer-arith -Wuninitialized -Wsign-compare -Wformat-security -Wsizeof-pointer-memaccess -Wno-pointer-
mplicit-fallthrough -Wmisleading-indentation -fno-strict-aliasing -D_FORTIFY_SOURCE=2 -ftwapi -fzero-call-used-regs=used -fno-builtin-memset -fstack-proted
Preprocessor flags: -I/home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/openssl/include -I/home/alyssa/Eltex/Eltex_tasks/19_cross_
-D_XOPEN_SOURCE=600 -D_BSD_SOURCE -D_DEFAULT_SOURCE -D_GNU_SOURCE
Linker flags: -L/home/alyssa/Eltex/Eltex_tasks/19_Cross compilation/build/src/openssh-9.7p1/openssl/lib -L/home/alyssa/Eltex/Eltex_tasks/19_cross_co
/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/zlib/lib -L/home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7
xestack -fstack-protector-strong -pie
Libraries:
+for channels: -lcrypto -lz
```

ошибок нет
библиотеки и заголовочные нашли
ссылка на libcrypto и libz подключена
тулчейн ARM распознан

собираем:
make -j12
make install-nokeys - сгенерируем ключи уже в кему

```
ong -pie -lssh -lopenbsd-compat -lcrypto -lz
> make install-nokeys
Makefile:690: предупреждение: игнорирование предварительных условий при определении суффиксного правила
(cd openbsd-compat && make)
make[1]: вход в каталог /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/openbsd-compat
make[1]: Цель «all» не требует выполнения команд.
make[1]: выход из каталога /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/openbsd-compat
/usr/bin/mkdir -p /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/bin
/usr/bin/mkdir -p /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/sbin
/usr/bin/mkdir -p /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/share/man/man1
/usr/bin/mkdir -p /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/share/man/man5
/usr/bin/mkdir -p /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/share/man/man8
/usr/bin/mkdir -p /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/libexec
/usr/bin/mkdir -p -m 0755 /var/empty
/usr/bin/mkdir: невозможно создать каталог «/var/empty»: Отказано в доступе
make: *** [Makefile:394: install-files] Ошибка 1
```

openssh пытается создать /var/empty - стандартный chroot-директории для privilege separation (из-за нашего --with-privsep-user=nobody)
хочу всё локально, поэтому переделаем конфиг заменив
на--with-privsep-path=\$PWD/_install/var/empty
make distclean

```
rm -rf _install
```

ВЕРНАЯ ПОПЫТКА:

```
mkdir -p $PWD/_install/var/empty - создаем chroot-директорию для privilege separation
```

```
./configure --prefix=$PWD/_install --host=arm-linux-gnueabi --with-zlib=$PWD/zlib  
--with-ssl-dir=$PWD/openssl --with-privsep-path=$PWD/_install/var/empty --disable-strip
```

заново

```
make -j12
```

```
make install-nokeys
```

ошибок нет, хвост вывода:

```
/usr/bin/install -c -m 644 sshd_config.5.out /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/share/man/man5/sshd_config.5  
/usr/bin/install -c -m 644 ssh_config.5.out /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/share/man/man5/ssh_config.5  
/usr/bin/install -c -m 644 sshd.8.out /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/share/man/man8/sshd.8  
/usr/bin/install -c -m 644 sftp.1.out /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/share/man/man1/sftp.1  
/usr/bin/install -c -m 644 sftp-server.8.out /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/share/man/man8/sftp-server.8  
/usr/bin/install -c -m 644 ssh-keysign.8.out /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/share/man/man8/ssh-keysign.8  
/usr/bin/install -c -m 644 ssh-pkcs11-helper.8.out /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/share/man/man8/ssh-pkcs11-helper.8  
/usr/bin/install -c -m 644 ssh-sk-helper.8.out /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/share/man/man8/ssh-sk-helper.8  
/usr/bin/mkdir -p /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/etc
```

```
/usr/bin/mkdir -p /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install  
> cd _install  
> ls  
bin  etc  libexec  sbin  share  var  
> ls *  
bin:  
scp  sftp  ssh  ssh-add  ssh-agent  ssh-keygen  ssh-keyscan  
  
etc:  
moduli  ssh_config  sshd_config  
  
libexec:  
sftp-server  ssh-keysign  ssh-pkcs11-helper  ssh-sk-helper  
  
sbin:  
sshd  
  
share:  
man  
  
var:  
empty  
  
~/E/Eltex_tasks/19/b/s/openssh/_install develop *1 !12 ?30 >
```

bin: клиентские утилиты

sbin: сервер sshd

libexec: вспомогательные утилиты

etc: конфиги

var/empty: chroot для sshd

➤ readelf -d bin/ssh

Динамический раздел со смещением 0x830 содержит 30 элементов:

Тег	Тип	Имя/Знач
0x00000001 (NEEDED)		Совм. исп. библиотека: [libcrypto.so.1.1]
0x00000001 (NEEDED)		Совм. исп. библиотека: [libz.so.1]
0x00000001 (NEEDED)		Совм. исп. библиотека: [libc.so.6]
0x00000001 (NEEDED)		Совм. исп. библиотека: [ld-linux-armhf.so.3]

создала lib в _install бибзипокса

добавила туда библиотеки из zlib и openssl (есть даже лишние, но пусть)

мне не хватает [libc.so.6] и [ld-linux-armhf.so.3]

они тут: /usr/arm-linux-gnueabi/lib

cp /usr/arm-linux-gnueabi/lib/ld-linux-armhf.so.3 .

➤ cp /usr/arm-linux-gnueabi/lib/libc.so.6 .

➤ cp /usr/arm-linux-gnueabi/lib/libc.so .

в итоге:

ls

ld-linux-armhf.so.3 libcrypto.so.1 libc.so libssl.so libz.so libz.so.1.3.1

libcrypto.so libcrypto.so.1.1 libc.so.6 libssl.so.1.1 libz.so.1

Теперь надо добавить openssh в _install бибзипокса

бинарники:

cp

/home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/bin/*
bin/

cp

/home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/sbin/*
sbin/

mkdir -p libexec

cp

/home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/libexec/*
libexec/

конфиги sshd: mkdir -p etc/ssh

cp

/home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/etc/*
etc/ssh/

добавляем в конфиг etc/ssh/sshd_config

Port 22

ListenAddress 0.0.0.0

HostKey /etc/ssh/ssh_host_rsa_key

HostKey /etc/ssh/ssh_host_ecdsa_key

```
HostKey /etc/ssh/ssh_host_ed25519_key
AuthorizedKeysFile .ssh/authorized_keys
PermitRootLogin yes
PermitEmptyPasswords yes
PasswordAuthentication yes
UsePrivilegeSeparation sandbox
UsePAM no
Subsystem sftp /libexec/sftp-server
```

```
Port 22
#AddressFamily any
ListenAddress 0.0.0.0
#ListenAddress ::

# HostKey /etc/ssh/ssh_host_rsa_key
# HostKey /etc/ssh/ssh_host_ecdsa_key
HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_
# but this is overridden so installations will
AuthorizedKeysFile .ssh/authorized_keys
```

```

# To disable tunneled clear text passwords, change
PasswordAuthentication yes
PermitEmptyPasswords yes

# Change to no to disable s/key passwords
#KbdInteractiveAuthentication yes

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no

# GSSAPI options
#GSSAPIAuthentication no
#GSSAPICleanupCredentials yes

# Set this to 'yes' to enable PAM authentication
# and session processing. If this is enabled, P
# be allowed through the KbdInteractiveAuthenti
# PasswordAuthentication. Depending on your PA
# PAM authentication via KbdInteractiveAuthenti
# the setting of "PermitRootLogin prohibit-pass
# If you just want the PAM account and session
# PAM authentication, then enable this but set
# and KbdInteractiveAuthentication to 'no'.
UsePrivilegeSeparation sandbox
UsePAM no

#Banner none

# override default of no subsystems
Subsystem      sftp      /libexec/sftp-server

```

Создаем root пользователя:

```

cat > etc/passwd <<EOF
root:x:0:0:root:/root:/bin/sh
EOF

```

```

cat > etc/group <<EOF
root:x:0:
EOF
mkdir -p root

```

Генерим ключи

```

ssh-keygen -t rsa -f etc/ssh/ssh_host_rsa_key -N ""

```

Generating public/private rsa key pair.

Your identification has been saved in etc/ssh/ssh_host_rsa_key

Your public key has been saved in etc/ssh/ssh_host_rsa_key.pub

The key fingerprint is:

SHA256:SGKHNw9P5e89AB+jjRCx7VgGsYfqmduZouG6oUI3YxE

alyssa@alyssa-BMH-WCX9

The key's randomart image is:

```
+---[RSA 3072]-----+
|      =o.      |
| E . X        |
|  = * * B o    |
| o = O B B o   |
| . o S + =     |
| . = . o . o   |
|. + + + . o    |
|.. o ..o o .   |
|o ooo...+      |
+----[SHA256]-----+
```

Сделаем init скрипт, где смонтируем виртуальные фс
запустим sshd

откроем интерактивную оболочку

```
#!/bin/sh
mount -t proc none /proc
mount -t sysfs none /sys
mount -t devtmpfs none /dev

echo "Boot OK"

/usr/sbin/sshd -f /etc/ssh/sshd_config
exec /bin/sh
```

chmod +x init

find . -print0 | cpio --null -ov --format=newc | gzip -9 > ../../initramfs.cpio.gz

```
./sbin/mkswap
./sbin/findfs
./sbin/pivot_root
23547 блоков
~/E/Eltex_tasks/18/b/_install develop *1 !12 ?
```

QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -m 256M -kernel zImage -dtb vexpress-v2p-ca9.dtb -initrd initramfs.cpio.gz -append "console=ttyAMA0 rdinit=/bin/ash" -nographic -net nic -net user,hostfwd=tcp::2222-:22 (256мб оперативки хватит, создаем виртуальный nic, NAT + проброс)

```
ALSA device list:
#0: ARM AC'97 Interface PL041 rev0 at 0x10004000, irq 41
Freeing unused kernel memory: 1024K
Run /bin/ash as init process
/bin/ash: can't access tty; job control turned off
~ #
~ # ssh -p 2222 root@localhost
Couldn't open /dev/null: No such file or directory
~ # ssh -p 2222 root@localhost
Couldn't open /dev/null: No such file or directory
~ # ls
bin      etc      lib      root     usr
dev      init     libexec  sbin
~ # cd dev
/dev # ls
console
/dev #
```

у меня нет стандартных устройств, которые требует опенсшш
создала

```
#0: ARM AC'97 Interface PL041 rev0 at 0x10004000, irq 41
Freeing unused kernel memory: 1024K
Run /bin/ash as init process
/bin/ash: can't access tty; job control turned off
~ #
~ # cd dev
/dev # mknod null c 1 3
/dev # mknod zero c 1 5
/dev # mknod tty c 5 0
/dev # mknod urandom c 1 9
/dev # chmod 666 null zero urandom
/dev #
```

```
crw-rw-rw- 1 root root 1, 5 Dec 1 16:14 zer
/dev # chmod 666 /dev/tty
/dev #
/dev # ls -l /dev
total 0
crw----- 1 root root 5, 1 Dec 1 22:02 con
crw-rw-rw- 1 root root 1, 3 Dec 1 16:07 nul
crw-rw-rw- 1 root root 5, 0 Dec 1 16:18 tty
crw-rw-rw- 1 root root 1, 9 Dec 1 16:23 ura
crw-rw-rw- 1 root root 1, 5 Dec 1 16:14 zer
/dev #
```

```
~ # /sbin/sshd -f /etc/ssh/sshd_config
/etc/ssh/sshd_config line 82: Deprecated option UsePrivilegeSeparation
/etc/ssh/sshd_config line 83: Unsupported option UsePAM
Privilege separation user sshd does not exist
~ #
```

удалить надо из конфига UsePrivilegeSeparation и Unsupported option UsePAM
создадим минимального юзера sshd

```
# echo "sshd:x:1:1:SSH Daemon:/var/empty:/bin/false" >> /etc/passwd
```

```
~ # mkdir -p /var/empty
```

```
~ # chmod 755 /var/empty
```

```
~ # /sbin/sshd -f /etc/ssh/sshd_config
```

```
Unable to load host key: /etc/ssh/ssh_host_ecdsa_key
```

```
Unable to load host key: /etc/ssh/ssh_host_ed25519_key
```

```
Missing privilege separation directory:
```

```
/home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/var/empty
```

хотела сгенерить ключи и создать симлинк, но пути захардкожены

```
~ # ssh-keygen -A
ssh-keygen: generating new host keys: RSA Could not save your private key in /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/etc/ssh_host_rsa_key.XXXX2S2FHA: No such file or directory
ssh-keygen: generating new host keys: ECDSA Could not save your private key in /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/etc/ssh_host_ecdsa_key.XXXXFoQEza: No such file or directory
ssh-keygen: generating new host keys: ED25519 Could not save your private key in /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/etc/ssh_host_ed25519_key.XXXXq6DyvT: No such file or directory
~ # mkdir -p /var/empty
~ # chmod 755 /var/empty
~ #
~ # ln -s /var/empty /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/var/empty
ln: /home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/var/empty: No such file or directory
~ #
```

идем исправлять

в опенсш

make clean

```
./configure --host=arm-linux-gnueabi --prefix=
```

```
--with-privsep-path=/var/empty \
```

```
--with-ssl-dir=$PWD/openssl \
```

```
--with-zlib=$PWD/zlib \
```

```
--disable-strip
```

```
make -j$(nproc)
make install-nokeys DESTDIR=$PWD/_install
```

```
чтобы перезаписать старые файлы новыми просто добавляем флаг -f
cp -f
/home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/bin/*
bin/
cp -f
/home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/sbin/*
sbin/
cp -rf
/home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/libexec/*
libexec/
cp -rf
/home/alyssa/Eltex/Eltex_tasks/19_cross_compilation/build/src/openssh-9.7p1/_install/etc/*
etc/ssh/
```

```
find . | cpio -o -H newc | gzip > ../../initramfs.cpio.gz
```

```
QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -dtb
vexpress-v2p-ca9.dtb -initrd initramfs.cpio.gz -append "console=ttyAMA0 rdinit=/bin/ash"
-nographic
```

Повтор шагов :

```
mkdir -p /var/empty
chmod 755 /var/empty
cd dev
mknod null c 1 3
mknod zero c 1 5
mknod urandom c 1 9
mknod tty c 5 0
/dev #
chmod 666 null zero urandom
chmod 600 tty
ssh-keygen -A
random: crng init done
ssh-keygen: generating new host keys: RSA ECDSA ED25519
cd ..
~ # echo "sshd:x:1:1:SSH Daemon:/var/empty:/bin/false" >> /etc/passwd
```

запускаем

```
~ # /sbin/sshd -f /etc/ssh/sshd_config
```

проверяем

```
~ # ssh -p 2222 root@localhost
```

ssh: Could not resolve hostname localhost: Temporary failure in name resolution

Уже что-то. ошибка в dns.

а блин я забыла при запуске в кему настройки сети задать

```
ssh -p 2222 root@127.0.0.1
```

ssh: connect to host 127.0.0.1 port 2222: Network is unreachable

короче

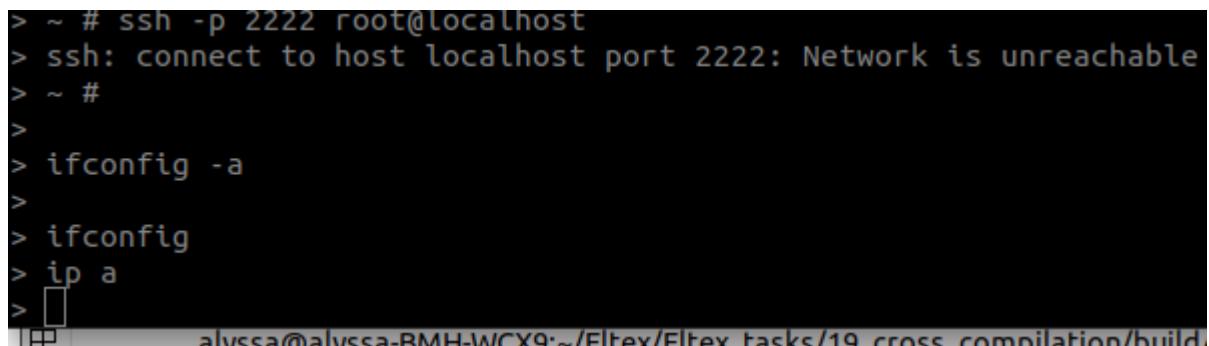
```
qemu-system-arm -M vexpress-a9 -kernel zImage -dtb vexpress-v2p-ca9.dtb -initrd
```

```
initramfs.cpio.gz -append "console=ttyAMA0 rdinit=/bin/ash" -netdev
```

```
user,id=net0,hostfwd=tcp::2222-:22 -device virtio-net-device,netdev=net0 -nographic
```

делаем Повтор шагов

сети нет



```
> ~ # ssh -p 2222 root@localhost
> ssh: connect to host localhost port 2222: Network is unreachable
> ~ #
>
> ifconfig -a
>
> ifconfig
> ip a
> [ ]
```

QEMU_AUDIO_DRV=none qemu-system-arm \

-M vexpress-a9 \

-kernel zImage \

-dtb vexpress-v2p-ca9.dtb \

-initrd initramfs.cpio.gz \

-append "console=ttyAMA0 rdinit=/bin/ash" \

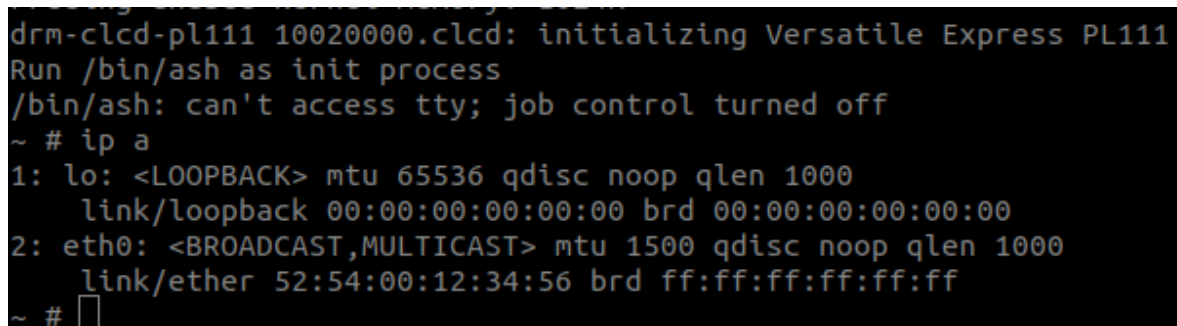
-netdev user,id=net0,hostfwd=tcp::2222-:22,dns=8.8.8.8 \

-device virtio-net-device,netdev=net0 \

-nographic

на vexpress virtio-net-device работает

отлично , сеть есть



```
drm-clcd-pl111 10020000.clcd: initializing Versatile Express PL111
Run /bin/ash as init process
/bin/ash: can't access tty; job control turned off
~ # ip a
1: lo: <LOOPBACK> mtu 65536 qdisc noop qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST> mtu 1500 qdisc noop qlen 1000
    link/ether 52:54:00:12:34:56 brd ff:ff:ff:ff:ff:ff
~ # [ ]
```

поднимем интерфейс `ip link set eth0 up`

```
~ # ip link set eth0 up
~ # udhcpd -i eth0
udhcpd: started, v1.36.1
udhcpd: broadcasting discover
udhcpd: broadcasting select for 10.0.2.15, server 10.0.2.2
udhcpd: lease of 10.0.2.15 obtained from 10.0.2.2, lease time 86400
~ #
```

DHCP не поднялся - назначаем вручную

`ip addr add 10.0.2.15/24 dev eth0`

`ip route add default via 10.0.2.2`

```
~ # ip a
1: lo: <LOOPBACK> mtu 65536 qdisc noop qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast
    link/ether 52:54:00:12:34:56 brd ff:ff:ff:ff:ff:ff
~ # ip addr add 10.0.2.15/24 dev eth0
~ # ip route add default via 10.0.2.2
~ # echo "nameserver 8.8.8.8" > /etc/resolv.conf
~ #
~ # ip a
1: lo: <LOOPBACK> mtu 65536 qdisc noop qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast
    link/ether 52:54:00:12:34:56 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 scope global eth0
        valid_lft forever preferred_lft forever
~ #
```

добавим в конфиг разрешения имен `echo "nameserver 8.8.8.8" > /etc/resolv.conf`

делаем Повтор шагов

`echo "root:x:0:0:root:/root:/bin/ash" >> /etc/passwd`

`echo "sshd:x:1:1:SSH Daemon:/var/empty:/bin/false" >> /etc/passwd`

`echo "sshd:x:1:" >> /etc/group`

пинги идут

```
64 bytes from 8.8.8.8: seq=4457 ttl=255 time=37.079 ms
64 bytes from 8.8.8.8: seq=4458 ttl=255 time=32.654 ms
64 bytes from 8.8.8.8: seq=4459 ttl=255 time=33.600 ms
64 bytes from 8.8.8.8: seq=4460 ttl=255 time=30.762 ms
64 bytes from 8.8.8.8: seq=4461 ttl=255 time=60.669 ms
64 bytes from 8.8.8.8: seq=4462 ttl=255 time=57.006 ms
64 bytes from 8.8.8.8: seq=4463 ttl=255 time=30.883 ms
64 bytes from 8.8.8.8: seq=4464 ttl=255 time=30.823 ms
64 bytes from 8.8.8.8: seq=4465 ttl=255 time=31.677 ms
64 bytes from 8.8.8.8: seq=4466 ttl=255 time=30.976 ms
64 bytes from 8.8.8.8: seq=4467 ttl=255 time=39.826 ms
64 bytes from 8.8.8.8: seq=4468 ttl=255 time=30.579 ms
64 bytes from 8.8.8.8: seq=4469 ttl=255 time=40.650 ms
64 bytes from 8.8.8.8: seq=4470 ttl=255 time=41.352 ms
64 bytes from 8.8.8.8: seq=4471 ttl=255 time=38.818 ms
64 bytes from 8.8.8.8: seq=4472 ttl=255 time=53.802 ms
QEMU: Terminated
```

```
passwd
Changing password for root
New password:
Retype password:
passwd: password for root changed by root
~ #
~ # mount -t proc proc /proc
~ # mount -t sysfs sysfs /sys
mount: mounting sysfs on /sys failed: No such file or directory
~ # mount -t devtmpfs devtmpfs /dev
~ # ??
/bin/ash: ды: not found
~ # ls
bin    etc    lib    proc   sbin   var
dev    init   libexec  root   usr
~ # mkdir sys
~ # mount -t sysfs sysfs /sys
~ # ip link set eth0 up
~ # udhcpc -i eth0
udhcpc: started, v1.36.1
udhcpc: broadcasting discover
udhcpc: broadcasting select for 10.0.2.15, server 10.0.2.2
udhcpc: lease of 10.0.2.15 obtained from 10.0.2.2, lease time 86400
~ #
```

```
# no default banner path
#Banner none

# override default of no subsystems
Subsystem      sftp /libexec/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#      X11Forwarding no
#      AllowTcpForwarding no
```

запустила sshd

а на хосте

```
> ssh -p 2222 root@localhost

@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@    WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED!    @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
IT IS POSSIBLE THAT SOMEONE IS DOING SOMETHING NASTY!
Someone could be eavesdropping on you right now (man-in-the-middle attack)!
It is also possible that a host key has just been changed.
The fingerprint for the ED25519 key sent by the remote host is
SHA256:hGWqPXlsDb3Yfuaw5IydSZbdQSA0mLdsrAUjxCeWLOQ.
Please contact your system administrator.
Add correct host key in /home/alyssa/.ssh/known_hosts to get rid of this message.
Offending ED25519 key in /home/alyssa/.ssh/known_hosts:7
  remove with:
    ssh-keygen -f "/home/alyssa/.ssh/known_hosts" -R "[localhost]:2222"
Host key for [localhost]:2222 has changed and you have requested strict checking.
Host key verification failed.

~/E/Eltex_tasks/19/b/s/openssh/_install develop *1 !12 ?30 >
```

ура ssh работает, и они правы

старая запись о сервере на порту 2222 была удалена, а резервная копия сохранена

```
> ssh-keygen -f "/home/alyssa/.ssh/known_hosts" -R "[localhost]:2222"

# Host [localhost]:2222 found: line 7
/home/alyssa/.ssh/known_hosts updated.
Original contents retained as /home/alyssa/.ssh/known_hosts.old

~/E/Eltex_tasks/19/b/s/openssh/_install develop *1 !12 ?30 >
```

```
> ssh -p 2222 root@localhost

The authenticity of host '[localhost]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:hGWqpXlsDb3Yfuaw5IydSZbdQSA0mLdsrAUjxCeWLOQ.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[localhost]:2222' (ED25519) to the list of known hosts.
Connection closed by 127.0.0.1 port 2222

~/E/Eltex tasks/19/b/s/openssh/ install develop *1 !12 230 > ❌
```

Ура оно работает и даже не ругается!